Kingsley_AI: Dynamic Web Lab Generation for Core CSS Concepts

1. Prompts

Initial Box Model Prompt

"Act as a frontend web developer. Using your Canvas tool, Generate an interactive website that can be used for understanding the CSS Box Model and its relationship with the display property. The page must have: 1. Two div elements, 'Box 1' and 'Box 2', so I can see how they interact. 'Box 1' will be the one we control. 2. The CSS must use different background colors for the content area, the padding area, and the margin area of 'Box 1' (e.g., using background-clip: content-box). The border should be a solid line. 3. A control panel with: - Sliders to control the padding, margin, border-width, and width of 'Box 1'. - Labels next to the sliders that show the current pixel value. - A Dropdown (select) to change the display property of 'Box 1' to: block, inline-block, and inline. 4. JavaScript that listens to all sliders and the dropdown, and updates the CSS properties of 'Box 1' in real-time."

Refinement Prompt

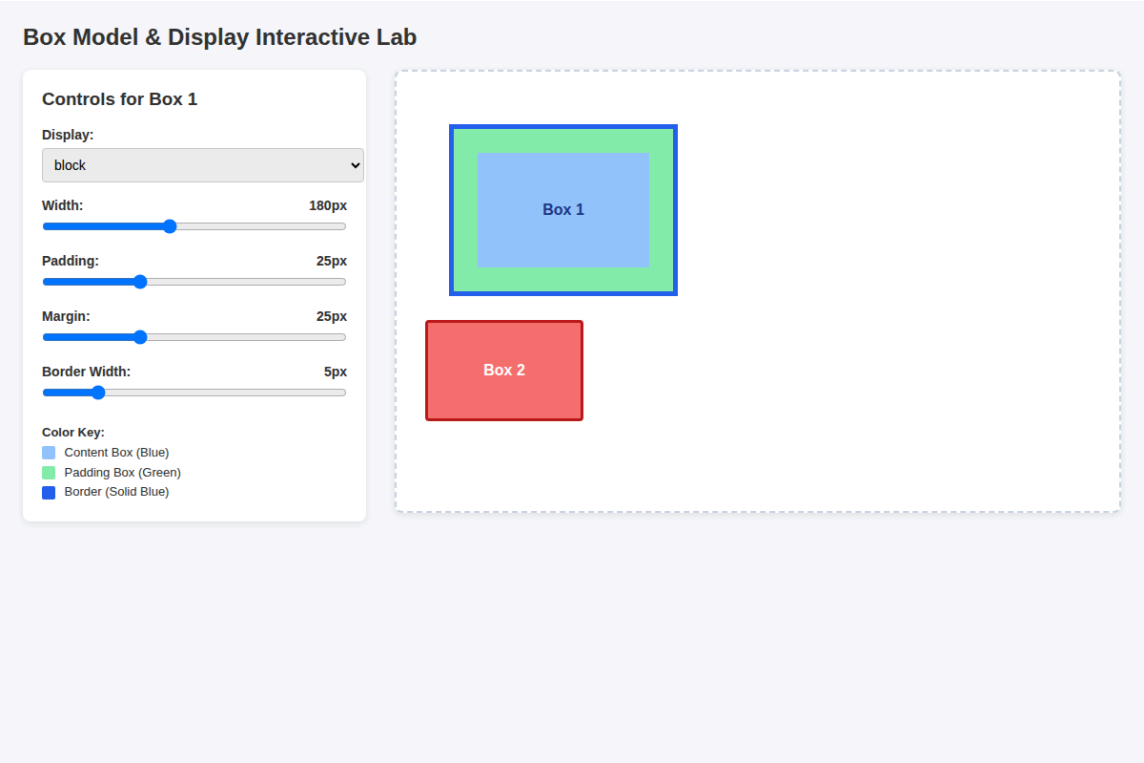
"Implement sliders to adjust the margin, padding, and border for each side (top, right, bottom, left) individually, and add a separate slider for the corner radius."

Flexbox and Grid Playground Prompt

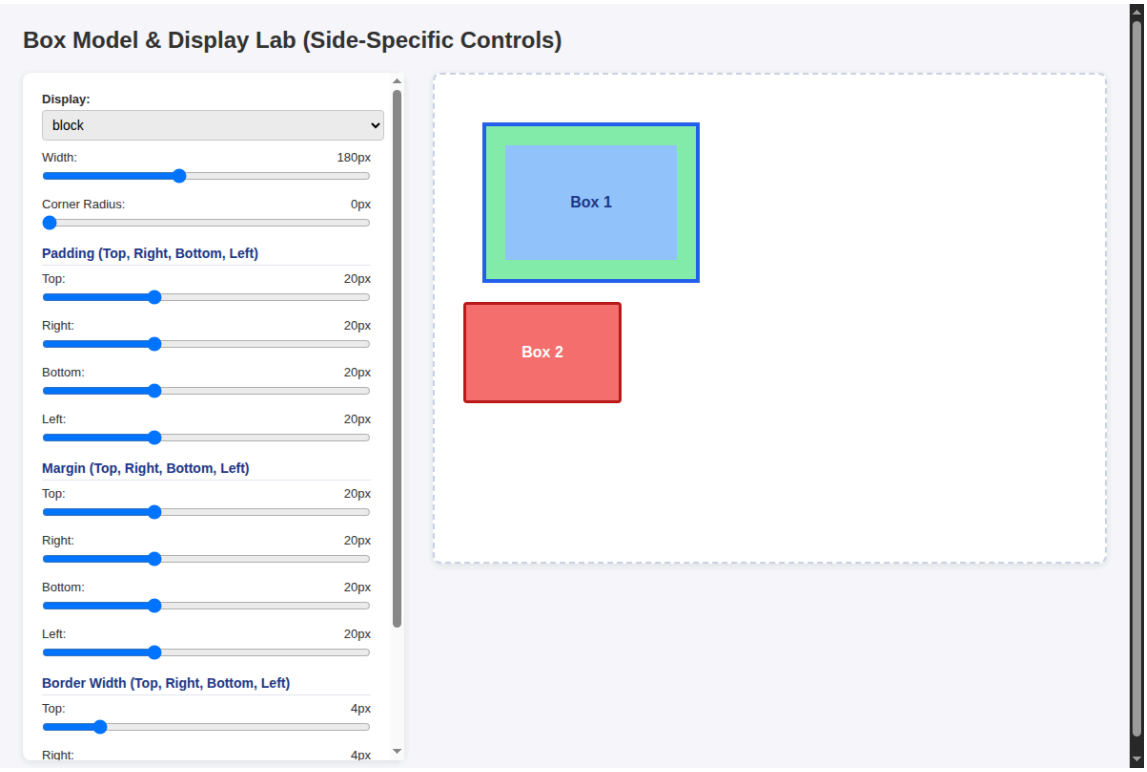
"Act as a frontend web developer. Using your Canvas tool, generate an interactive website that can be used as a playground for CSS Flexbox and Grid. The page should have: 1. A div element acting as the container. 2. Several div elements inside acting as the items (e.g., 5 items). 3. Dropdown menus (selects) that allow me to change the CSS properties of the container. 4. I need to be able to change: - display (to switch between block, flex, and grid) - flex-direction (row, column) - justify-content (flex-start, center, space-between, etc.) - align-items (flex-start, center, stretch, etc.) - grid-template-columns (e.g., 1fr 1fr, 1fr 1fr 1fr) 5. The JavaScript must update the container's CSS in real-time when I change a dropdown."

2. Screenshots

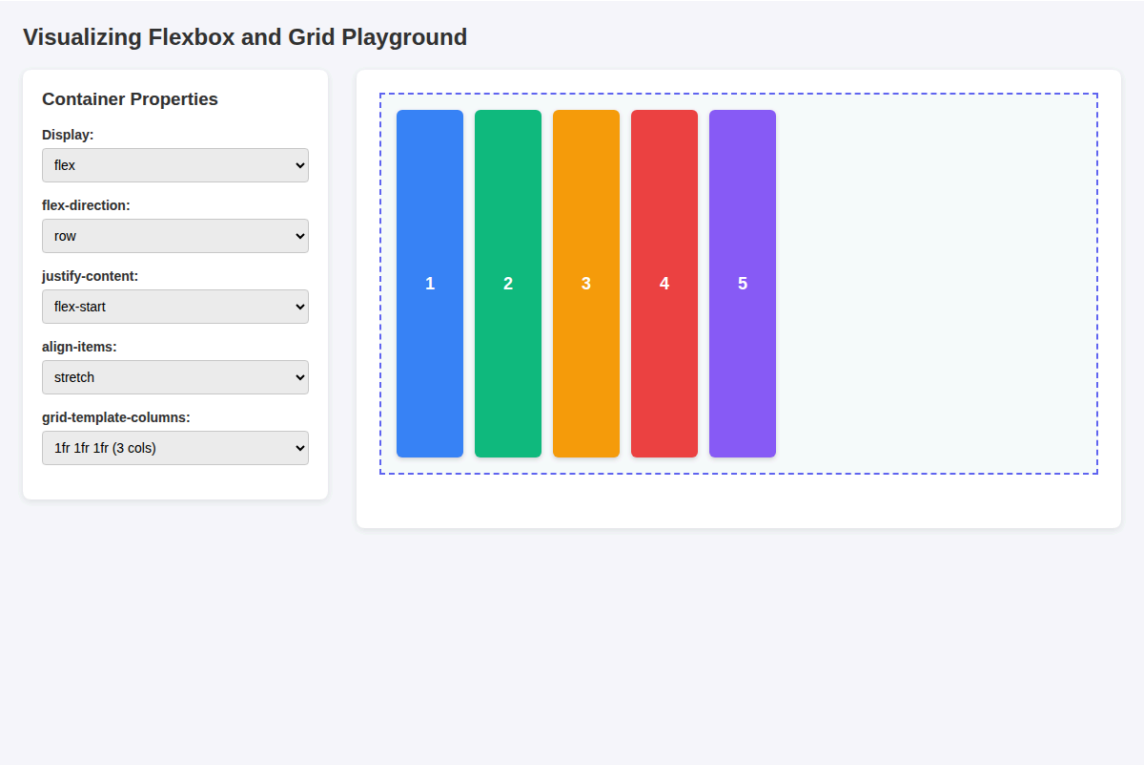
- Screenshot 1: Initial Box Model Lab Output



- Screenshot 2: Refined Box Model Lab with Side-Specific Controls Output



- Screenshot 3: Flexbox & Grid Playground Output



3. Reflection and Synthesis

Using interactive, live-updating web tools completely transforms how CSS concepts are understood compared to reading static documentation or looking at static diagrams. Reading about properties like padding, margin, or display in text often feels abstract, making it hard to predict how layout rules interact in practice. For instance, when switching Box 1 to display: inline, seeing in real-time how setting explicit width values or top and bottom margins no longer affects vertical layout immediately brings the rule to life. That instant visual feedback loop bridged the gap between theory and actual browser rendering, making core CSS behaviors click much faster than standard documentation ever did.

From an AI workflow perspective, building the lab incrementally through follow-up refinement prompts proved to be much more reliable and manageable than trying to request the complete, complex application in a single prompt. Starting with a minimal working Box Model tool allowed for verifying core functionality before introducing side-specific controls or corner radius options. Attempting to prompt every feature at once often results in missed requirements or hard-to-debug code. Iterating step-by-step closely mirrors real-world agile software development, where features are built, tested, and expanded in focused cycles to maintain quality and control.

4. GitHub Repository Link

GitHub Repository: AI: Dynamic Web Lab Generation for Core CSS Concepts